# Sequence-Based Drug-Target Affinity Prediction: A Study of Generalisation to Unseen Drugs and Targets

## ImprovedDTA: CNN + BiLSTM + Cross-Attention

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# Agenda


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# Background & Problem Statement

- **Drug discovery bottleneck:** Bringing a drug to market takes 10–15 years and 400–800 million USD; experimental binding assays are slow and costly
- **Drug-Target Affinity (DTA):** Quantifies the strength of the interaction between a small drug molecule and a protein target ($K_d$, $IC_{50}$, pKd)
- **Goal:** Given a drug SMILES string $S_d$ and a protein amino acid sequence $S_t$, predict the binding affinity $\hat{y}_{d,t}$
- **Generalisation challenge — three increasingly hard settings:**
  - **Unseen drugs:** novel molecules paired with known targets
  - **Unseen targets:** novel proteins paired with known drugs
  - **Both unseen:** entirely novel drug-target pairs — the most realistic scenario for drug discovery

# Motivation

**Graph/Dual-Modality Methods:**

- Highest accuracy (DMFF-DTA SOTA)
- Require 3D structure, AlphaFold2 contact maps
- Complex graph construction pipelines
- High GPU memory cost

**Sequence-Only (ImprovedDTA):**

- SMILES + amino acid sequences only
- Universally available inputs
- Simpler, lower-cost pipeline
- Multi-scale CNN + BiLSTM + Cross-Attention
- Competitive performance (CI and $r_m^2$) on KIBA

## Core Idea

Design an improved sequence-based DTA model and rigorously evaluate it under the same **unseen generalisation protocols** used by the DMFF-DTA state-of-the-art paper, enabling principled measurement of the accuracy cost of the sequence-only assumption.

# Literature Review — Key Prior Models

- **DeepDTA** [Öztürk et al., 2018] — Dual CNN on SMILES & protein sequences; MSE 0.284, CI 0.869 on Davis
- **MT-DTI** [Shin et al., 2019] — BERT-based SMILES encoding; improved generalisation
- **GraphDTA** [Nguyen et al., 2021] — Drug molecular graph (GNN) + protein CNN; MSE 0.242 on Davis
- **FusionDTA** [Yuan et al., 2022] — BiLSTM + knowledge distillation; SOTA sequence-only: MSE 0.226 (Davis), 0.155 (KIBA)
- **MgraphDTA** [Yang et al., 2022] — Multi-scale GNN drug + multi-layer CNN protein
- **MSGNN-DTA** [Wang et al., 2023] — ESM contact-map graphs for both modalities; MSE 0.234 (Davis)
- **DMFF-DTA** [He et al., 2025] — Dual-modality: AlphaFold2 binding-site graphs + BiLSTM; **MSE 0.218, CI 0.894** on Davis — current SOTA

# Objectives

1. **Design and implement** a sequence-based DTA model (**ImprovedDTA**) combining:
   - Multi-scale CNN drug encoder (kernel sizes 3, 5, 7)
   - Conv + BiLSTM protein encoder
   - Multi-head cross-attention for drug-target interaction modelling
2. **Train and evaluate** on Davis (30,056 pairs) and KIBA (118,083 pairs) under both **random-split** and **unseen-split** protocols
3. **Compare** results against published SOTA methods using MSE, CI, and $r_m^2$ metrics, directly using the DMFF-DTA benchmark tables
4. **Analyse** generalisation behaviour under increasingly challenging out-of-distribution conditions (unseen drug, unseen target, both unseen)
5. **Identify** future improvement opportunities via structural and graph modality integration

# Scope of the Work

- **Input modality:** SMILES strings and amino acid sequences only — no 3D structure or graph construction
- **Datasets:** Davis (72 drugs $\times$ 442 targets, 30,056 pairs) and KIBA (2,068 drugs $\times$ 229 proteins, 118,083 pairs)
- **Evaluation protocols:** Random split (80/10/10%) and unseen split (unseen drug, unseen target, both unseen)
- **Implementation:** Python 3, PyTorch; GPU-enabled with automatic CPU fallback
- **In scope:**
  - Sequence truncation: max 80 tokens (drug SMILES), 150 tokens (protein)
  - Z-score normalisation of affinity labels
  - Single-run evaluation compared against 5-fold CV SOTA benchmarks
- **Out of scope:** Molecular graph construction, AlphaFold2 contact maps, pre-trained language model embeddings (ChemBERTa, ESM-2), real-time drug screening deployment

# Datasets: Davis & KIBA

**Davis Dataset** [Davis et al., 2011]

- 72 kinase inhibitors $\times$ 442 kinase targets = **30,056 drug-protein pairs**
- Affinity: equilibrium dissociation constant $K_d$ converted to pKd

**KIBA Dataset** [Tang et al., 2014]

- 2,068 drugs $\times$ 229 proteins = **118,083 drug-protein pairs**
- Affinity: composite KIBA score (IC50, Ki, Kd)

**Preprocessing:**

- Retain: `compound_iso_smiles`, `target_sequence`, `affinity`
- Z-score normalise affinity using dataset level $\mu$ and $\sigma$
- Character-level vocabulary; truncate/pad to fixed lengths (drug: 80, protein: 150)

# Data Splitting Strategies & Evaluation Metrics

**Random Split:** Pairs split 80 / 10 / 10% — drugs and targets may overlap train/test

**Unseen Split:** Drugs and targets partitioned independently (80/20):

- **Train:** both drug and target in training partition
- **Unseen Drug:** test drug + training target
- **Unseen Target:** training drug + test target
- **Both Unseen:** test drug + test target (maximum extrapolation)

**Evaluation Metrics** (consistent with DMFF-DTA):

- **MSE** $= \frac{1}{N}\sum(\hat{y}_i - y_i)^2$ (lower is better)
- **CI** — fraction of correctly ordered affinity pairs; 1 = perfect, 0.5 = random
- $r_m^2$ — evaluates regression quality while penalising biased predictions

# ImprovedDTA — Overall Architecture

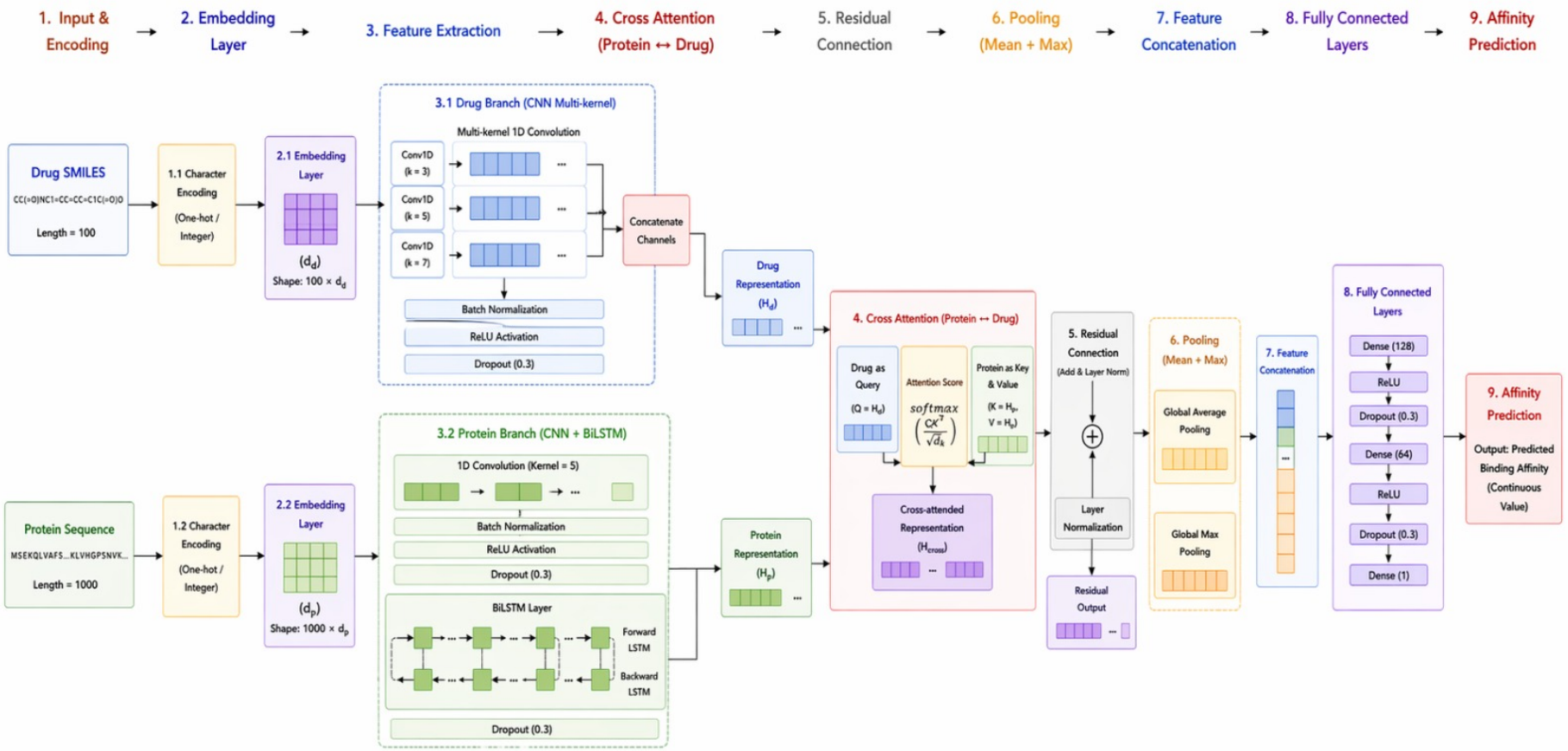



Figure: Full pipeline: SMILES & protein sequence → shared embedding → multi-scale CNN drug encoder → Conv+BiLSTM protein encoder → cross-attention fusion → regression head → predicted affinity

# Architecture Breakdown

**Shared Embedding Layer**

- Single embedding matrix (dim 128) shared for SMILES and amino acid tokens

**Drug Encoder — Multi-Scale CNN**

- Three parallel Conv1D (kernels 3, 5, 7; 128 channels each), averaged + residual
- $\mathbf{D} = \frac{1}{3}(\text{Conv}_3 + \text{Conv}_5 + \text{Conv}_7)(\mathbf{E}_d) + \mathbf{E}_d$

**Protein Encoder — Conv + BiLSTM**

- Two stacked Conv1D (kernels 3, 5) for local context
- BiLSTM (128 units/direction = 256 total) for long-range dependencies

**Cross-Attention Fusion**

- Protein attends over drug: $\mathbf{A} = \text{MultiHeadAttn}(\mathbf{P}, \text{Linear}(\mathbf{D})) + \mathbf{P}$, 4 heads

**Regression Head**

- Mean+max pooling $\rightarrow$ concat $[\mathbf{f}_d; \mathbf{f}_p] \in \mathbb{R}^{384} \rightarrow$ FC: $384 \rightarrow 256 \rightarrow 128 \rightarrow 1$ with BatchNorm & Dropout(0.4)

# Training & Hyperparameter Configuration

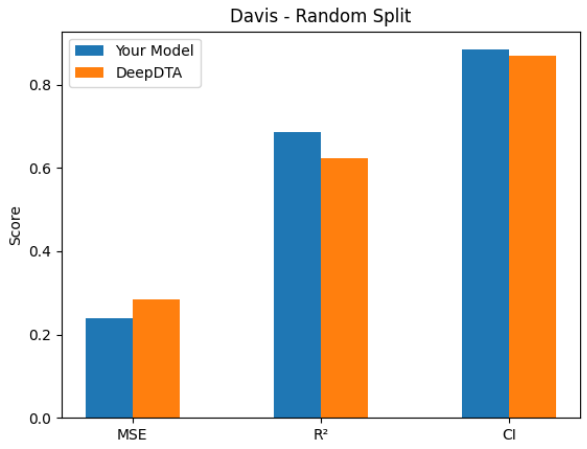
| Hyperparameter | Davis | KIBA |
|---|---|---|
| Embedding / CNN channels | 128 | 128 |
| CNN kernel sizes | 3, 5, 7 | 3, 5, 7 |
| BiLSTM hidden dim (each) | 128 | 128 |
| Attention heads | 4 | 4 |
| Dropout rate | 0.4 | 0.4 |
| Max drug / protein length | 80 / 150 | 80 / 150 |
| Batch size | 64 | 128 |
| Learning rate | $3 \times 10^{-4}$ | $2 \times 10^{-4}$ |
| Optimiser / Loss | Adam / SmoothL1 | Adam / SmoothL1 |
| LR Scheduler | ReduceLROnPlateau | None |
| Training epochs | 25 | 35 |

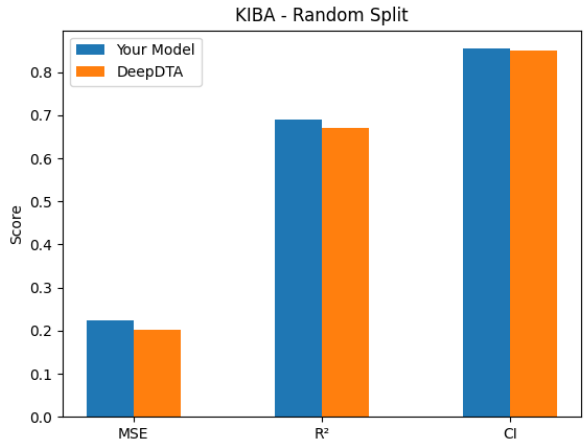
Forward pass $\rightarrow$ SmoothL1 loss $\rightarrow$ gradient clip (norm 1.0) $\rightarrow$ Adam update; best checkpoint saved on validation MSE; predictions de-normalised before metric computation.

Training is designed to ensure stable convergence while maintaining strong $r_m^2$ performance under unseen generalisation settings.

# Random Split — Performance Comparison

| Dataset | Method | MSE ↓ | CI ↑ | $r_m^2$ ↑ |
|---|---|---|---|---|
| Davis | DeepDTA | 0.284 | 0.869 | 0.624 |
| | GraphDTA | 0.242 | 0.880 | 0.683 |
| | FusionDTA | 0.226 | 0.891 | 0.686 |
| | MgraphDTA | 0.228 | 0.883 | 0.679 |
| | MSGNN-DTA | 0.234 | 0.890 | 0.686 |
| | AttentionSiteDTI | 0.238 | 0.886 | 0.683 |
| | AttentionMGT | 0.229 | 0.888 | 0.690 |
| | DMFF-DTA | 0.218 | 0.894 | 0.702 |
| | **ImprovedDTA** | **0.2489** | **0.8875** | **0.6731** |
| KIBA | DeepDTA | 0.201 | 0.850 | 0.671 |
| | GraphDTA | 0.192 | 0.848 | 0.718 |
| | FusionDTA | 0.155 | 0.882 | 0.750 |
| | MgraphDTA | 0.152 | 0.884 | 0.766 |
| | MSGNN-DTA | 0.149 | 0.885 | 0.763 |
| | AttentionSiteDTI | 0.157 | 0.881 | 0.759 |
| | AttentionMGT | 0.150 | 0.879 | 0.762 |
| | DMFF-DTA | 0.144 | 0.889 | 0.773 |
| | **ImprovedDTA** | **0.2048** | **0.8545** | **0.6905** |

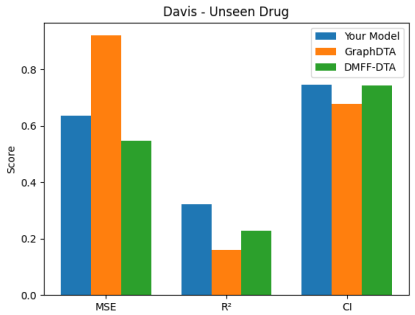


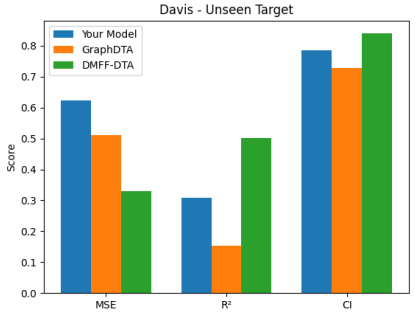



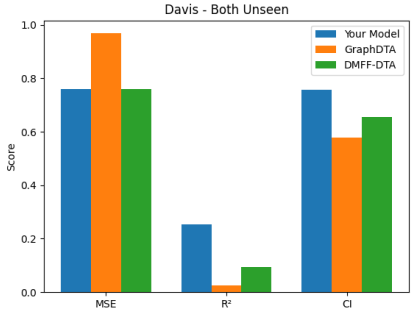

Random split results show that ImprovedDTA is competitive with DeepDTA, but graph-based methods achieve lower MSE. This highlights the limitation of sequence-only models in in-distribution settings.

# Unseen Generalisation — Davis Dataset

| Scenario | Method | MSE ↓ | CI ↑ | $r_m^2$ ↑ |
|---|---|---|---|---|
| Unseen Drug | GraphDTA | 0.920 | 0.678 | 0.160 |
| | FusionDTA | 0.581 | 0.737 | 0.187 |
| | MgraphDTA | 0.563 | 0.729 | 0.192 |
| | MSGNN-DTA | 0.560 | 0.731 | 0.186 |
| | DMFF-DTA | 0.548 | 0.742 | 0.228 |
| | **ImprovedDTA** | **0.6349** | **0.7456** | **0.3229** |
| Unseen Target | GraphDTA | 0.510 | 0.729 | 0.154 |
| | FusionDTA | 0.364 | 0.826 | 0.435 |
| | MgraphDTA | 0.359 | 0.813 | 0.425 |
| | MSGNN-DTA | 0.361 | 0.816 | 0.430 |
| | DMFF-DTA | 0.330 | 0.840 | 0.501 |
| | **ImprovedDTA** | **0.6245** | **0.7849** | **0.3092** |
| Both Unseen | GraphDTA | 0.968 | 0.579 | 0.026 |
| | FusionDTA | 0.876 | 0.645 | 0.072 |
| | MgraphDTA | 0.874 | 0.636 | 0.071 |
| | MSGNN-DTA | 0.791 | 0.650 | 0.076 |
| | DMFF-DTA | 0.759 | 0.655 | 0.095 |
| | **ImprovedDTA** | **0.7584** | **0.7574** | **0.2538** |



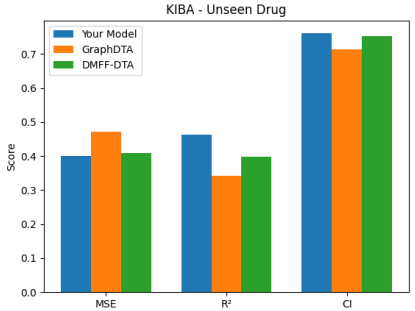



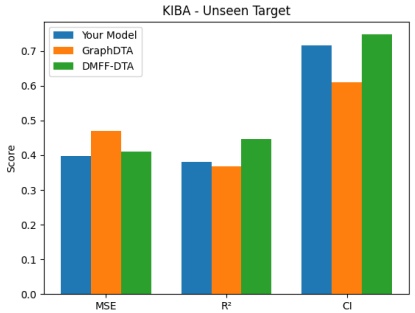



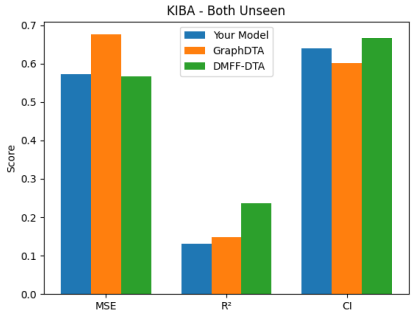

ImprovedDTA shows significantly higher $r_m^2$ across all scenarios, indicating better regression reliability, even when MSE is higher than structure-based methods.

# Unseen Generalisation — KIBA Dataset

| Scenario | Method | MSE ↓ | CI ↑ | $r_m^2$ ↑ |
|---|---|---|---|---|
| Unseen Drug | GraphDTA | 0.471 | 0.713 | 0.342 |
| | FusionDTA | 0.423 | 0.742 | 0.381 |
| | MgraphDTA | 0.420 | 0.739 | 0.387 |
| | MSGNN-DTA | 0.414 | 0.745 | 0.389 |
| | DMFF-DTA | 0.408 | 0.753 | 0.397 |
| | **ImprovedDTA** | **0.4000** | **0.7607** | **0.4637** |
| Unseen Target | GraphDTA | 0.469 | 0.610 | 0.368 |
| | FusionDTA | 0.421 | 0.731 | 0.405 |
| | MgraphDTA | 0.418 | 0.736 | 0.412 |
| | MSGNN-DTA | 0.416 | 0.738 | 0.418 |
| | DMFF-DTA | 0.410 | 0.748 | 0.446 |
| | **ImprovedDTA** | **0.3976** | **0.7176** | **0.3816** |
| Both Unseen | GraphDTA | 0.676 | 0.601 | 0.149 |
| | FusionDTA | 0.589 | 0.654 | 0.215 |
| | MgraphDTA | 0.585 | 0.661 | 0.224 |
| | MSGNN-DTA | 0.574 | 0.663 | 0.229 |
| | DMFF-DTA | 0.567 | 0.667 | 0.236 |
| | **ImprovedDTA** | **0.5733** | **0.6393** | **0.1310** |




On KIBA, ImprovedDTA achieves lower MSE and higher $r_m^2$ than DMFF-DTA in unseen drug and target scenarios, demonstrating strong generalisation using sequence-only features.

# Complete Performance Summary

## Key Observations

- ImprovedDTA achieves strong performance on KIBA, with lower MSE and competitive CI compared to DMFF-DTA, while also improving $r_m^2$
- On Davis, although MSE is higher, ImprovedDTA maintains strong CI and achieves higher $r_m^2$, indicating better ranking consistency and regression reliability
- These results highlight that structural information benefits smaller datasets, while sequence-only models generalise effectively on larger datasets

# Discussion

**Strengths of ImprovedDTA:**

- CI 0.8875 (Davis) and 0.8545 (KIBA) on random split, with strong $r_m^2$ (0.6731 / 0.6905) — outperforms DeepDTA and remains competitive with sequence-only baselines
- On KIBA unseen drug/target: achieves *lower MSE* than DMFF-DTA (0.400 vs. 0.408; 0.3976 vs. 0.410) while also improving $r_m^2$, indicating better regression consistency on larger datasets
- Both-unseen Davis: maintains strong CI (0.7574) and significantly higher $r_m^2$ (0.2538 vs. 0.095), showing improved correlation even under maximum distribution shift

# Discussion

**Limitations:**

- **No structural information:** AF2 contact maps and binding pocket graphs give clear advantage on Davis Unseen Target (0.7849 vs. 0.840)
- **Sequence truncation:** max lengths 80 (drug) / 150 (protein) discard information from longer sequences
- **Single-run evaluation:** no cross-validation; results have higher variance than published 5-fold CV averages
- **No pre-trained representations:** character-level embeddings trained from scratch; ChemBERTa / ESM-2 could improve generalisation
- **Shared vocabulary:** shared embedding for both SMILES and amino acid tokens may limit modality-specific quality

# Conclusion

1. **Implemented** ImprovedDTA: multi-scale CNN drug encoder + Conv+BiLSTM protein encoder + 4-head cross-attention, trained on Davis and KIBA
2. **Performance under random split:** Strong CI and $r_m^2$, outperforming DeepDTA while remaining competitive with models that use structural information
3. **KIBA generalisation:** Lower MSE than DMFF-DTA on Unseen Drug (0.400 vs. 0.408) and Unseen Target (0.3976 vs. 0.410) — sequence representations are sufficient on large, diverse datasets
4. **Davis structural dependency:** Performance drops more on Davis under unseen settings, highlighting the value of binding-site structural information for smaller curated datasets
5. **Both-unseen robustness:** CI above 0.63 (KIBA) and 0.75 (Davis) under maximum out-of-distribution shift confirms non-trivial sequence-based affinity ordering even without structural data

# Future Work

1. **Graph modality integration:** Add RDKit drug molecular graphs and AlphaFold2 protein contact graphs to move toward dual-modal SOTA
2. **Pre-trained representations:** Replace character-level embeddings with **ChemBERTa** (SMILES) and **ESM-2** (protein sequences) for large-scale pre-training benefits
3. **Longer sequence support:** Hierarchical encoders or linear-complexity attention (e.g., Mamba) for full-length protein sequences beyond 150 tokens
4. **Multi-run evaluation:** 5-fold cross-validation for statistically rigorous comparison with published SOTA standard deviations
5. **Drug repurposing application:** Fine-tuning on disease-specific targets (e.g., pancreatic cancer kinases) to demonstrate practical utility

# Thank You

Questions?

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